Fiscal Year:	FY 2018	**Task Last Updated:**	FY 05/14/2018
PI Name:	Kiss, John Z Ph.D.		
Project Title:	Novel Explorations Into The Interactions Between Light and Gravity Sensing in Plants		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	(1) Developmental Biology		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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PI Organization Type:	UNIVERSITY	**Phone:**	336-334-5241
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City:	Greensboro	**State:**	NC
Zip Code:	27402-6170	**Congressional District:**	13
Comments:	NOTE: PI moved to Florida Institute of Technology from University of North Carolina-Greensboro in May 2024 (Ed., 2/6/25). NOTE: PI moved to University of North Carolina-Greensboro from University of Mississippi in July 2016 (Ed., 6/13/17) NOTE: PI moved to U Miss from Miami (Ohio) University in September 2012 (Ed., 8/10/2012)		
Project Type:	Flight	**Solicitation / Funding Source:**	2009 Space Life Sciences-Flight NNH09ZTT002N
Start Date:	08/10/2017	**End Date:**	08/09/2019
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	1
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA ARC
Contact Monitor:	Sato, Kevin	**Contact Phone:**	650-604-1104
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Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	80NSSC17K0546		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	[ED. NOTE 12/11/2017: Continuation of "Novel Explorations Into the Interactions Between Light and Gravity Sensing in Plants--NNX12AO65G," per PI move from University of Mississippi to University of North Carolina at Greensboro. See that project for previous reports] The major goals of this project are to determine how gravity and light responses influence each other in plants and to better understand the cellular signaling mechanisms involved in plant tropisms. This proposed project builds on our previous spaceflight experiments on the International Space Station (ISS) with the European Modular Cultivation System (EMCS) using the model plant Arabidopsis. In this new proposed project, we plan to confirm and extend our discovery during the recent ISS experiment of a novel red-light-based positive phototropic response in plant shoots. Our hypothesis is that positive red-light-sensing, which was known in older plant lineages, is masked by normal 1-g conditions in more recently evolved lineages. We also propose to investigate fundamental interactions among red and blue light signaling pathways and the gravity sensing mechanisms. The ISS is the only laboratory facility where it is possible to reliably and simultaneously vary light and gravity stimuli for the proposed experiments. These experiments can be accomplished with telemetric science and therefore fit with the extremely limited down mass capabilities of current spaceflight opportunities. The EMCS is an automated facility and can be operated largely with telemetry. Experimental containers will launched via Space-X, and during an experimental run, images of seedling growth responses can be downlinked to Earth. A significant advantage of this proposed project is that we will use existing flight hardware that has been flown successfully and allowed for growth of Arabidopsis seedlings in the EMCS. This project is also relevant to the recent emphasis at NASA to study plant growth and development at fractional g-levels such as those found on the moon and Mars. Improved knowledge of the basic mechanistic processes that will be the focus of this project is vital to develop ways to use plants in extraterrestrial bioregenerative life support systems.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Studying the effects of gravity and light on basic cellular mechanisms in plants, this project is relevant to the emphasis on using Arabidopsis thaliana as a model system in developmental biology. Since plants will be a necessary part of bioregenerative life support needed to send humans to Mars and beyond, the knowledge obtained from our spaceflight experiments will be critical for developing ways to effectively use plants in the bioregenerative life support systems. These life support systems also can be used in ground-based Earth applications such as the design of large-scale greenhouses to support crop plants and agriculture. In addition, improved knowledge of light and gravity systems in plants will, in the long term, allow us to develop better crop plants for use on Earth.
Task Progress:	GENERAL SUMMARY OF ACTIVITIES: 1. Seedling Growth-3 space experiment was performed on the ISS during the SpaceX-11 mission in June 2017. 2. Samples were returned on SpaceX-11 in July 2017. These consisted of frozen and fixed seedlings. 3. We are in the process of cytological analyses of samples. 4. We also are performing RNAseq analysis, and the results will be deposited in NASA's GeneLab. 5. We also are doing image analysis of the samples to study growth, development, and tropistic curvature. SPECIFIC RESULTS FROM SPACEFLIGHT EXPERIMENT: Plants will play an important role in the future of space exploration as part of bioregenerative life support. Thus, it is important to understand the effects of microgravity and spaceflight on gene expression in plant development. We analyzed the transcriptome of Arabidopsis thaliana grown is space. The bioinformatics methods used included RMA (Robust Multi-Array Average), MAS5 (Microarray Suite 5.0), PLIER (Probe Logarithmic Intensity Error Estimation). Glycome profiling was utilized to analyze cell wall composition in the samples. In addition, our results were compared to those of two other groups using the same hardware on the same mission. In these space experiments, we noted expression changes in genes involved in hypoxia, heatshock, DNA repair, and cell wall structure between spaceflight samples compared to the ground controls. In addition, glycome profiling supported our expression analyses in that there was a difference in cell wall components between ground control and spaceflight-grown plants. A common theme from our space experiments and those of the other two groups was the down-regulation of water stress response genes in spaceflight. In addition, all three studies found differential regulation of genes associated with cell wall remodeling and stress responses between spaceflight-grown and ground control plants.
Bibliography Type:	Description: (Last Updated: 11/25/2025)
Abstracts for Journals and Proceedings	Medina FJ, Manzano-Pérez AMP, Ciska MC, Sáez-Vásquez JSV, Herranz R, Vandenbrink JP, Kiss JZ. "Role of light in the adaptation of plants to microgravity as shown by changes in gene expression after an ISS experiment." 7th ISPS & 25th ELGRA Biennial Symposium and General Assembly, Juan-les-Pins, France, October 2 -6, 2017. Abstract Book. 7th ISPS & 25th ELGRA Biennial Symposium and General Assembly, Juan-les-Pins, France, October 2 -6, 2017. , Oct-2017
Abstracts for Journals and Proceedings	Kiss JZ, Johnson CM, Edelmann RE, Medina FJ. "The limits of using the BRIC spaceflight hardware to study plant development." 7th ISPS & 25th ELGRA Biennial Symposium and General Assembly, Juan-les-Pins, France, October 2 -6, 2017. Abstract Book. 7th ISPS & 25th ELGRA Biennial Symposium and General Assembly, Juan-les-Pins, France, October 2 -6, 2017. , Oct-2017
Abstracts for Journals and Proceedings	Kiss JZ, Johnson CM, Edelmann RE, Correll MJ. "Comparative transcriptomics in spaceflight studies indicate changes in cell wall organization." 33rd Annual Meeting of the American Society for Gravitational and Space Research, Seattle, WA, October 25-28, 2017. 33rd Annual Meeting of the American Society for Gravitational and Space Research, Seattle, WA, October 25-28, 2017. Abstract Book, p. 42. , Oct-2017

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Articles in Peer-reviewed Journals	Johnson CM, Subramanian A, Pattathil S, Correll MJ, Kiss JZ. "Comparative transcriptomics indicate changes in cell wall organization and stress response in seedlings during spaceflight." American Journal of Botany. 2017 Aug;104(8):1219-31. https://doi.org/10.3732/ajb.1700079 ; PubMed PMID: 28827451; PubMed Central PMCID: PMC5821596 , Aug-2017
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Awards	Kiss JZ. (John Z. Kiss) "Founder's Award, American Society for Gravitational & Space Research. This award is the highest honor given by the Society for lifelong contributions. Awarded at October 2017 annual meeting, Seattle, WA, October 2017." Oct-2017
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